

Complex Engineering Problem

Course Code and Title: EE3004 – Feedback Control Systems

Semester: Spring 2025

Instructors: Dr. Omer Saleem & Ms. Tamania Javaid

Total Marks: 50

Deadline: **Monday, 13th May 2025**

CLOs and PLOs for Complex Engineering Problem

CLO		Description	Taxonomy Levels	PLO
CLO 5	Theory	Design a closed-loop feedback controller	C5	3

Instructions

Attempt all parts of the following design problem.

Problem Statement

Design a closed loop feedback control system for a customized transfer function $G(s)$. Considering your roll number is **22L – abcd**, your forward path transfer function would be

$$G(s) = \frac{(s + 22)}{(s + a)(s + b)(s + c + dj)(s + c - dj)}$$

Objectives

1. **Sketch** the root locus assuming it is a unity feedback control system. Show all the calculations related to asymptotes, break points, $j\omega$ crossings and angle of departure/arrival.
2. **Design** a PID controller for your plant such that the maximum overshoot of the overall system would be 12% and the system would have a settling time of 8.0 sec. There should be no error in step response. Show your complete step by step design theoretically.
3. **Design** the circuit of PID compensated system.
4. **Design** a lead-lag compensator using the transient specifications given in part (b) and 15% reduction in steady state error. Show your complete step by step design theoretically.
5. **Design** the circuit of lead-lag compensated system.
6. **Compare** the two controllers in terms of steady state and transient response parameters and suggest their advantages and disadvantages. Also recommend which one should be adopted for implementation.

Deliverables

- Schematic diagrams with detailed annotations and theoretical analysis for the controller design and evaluation of critical gains
- Design Report discussing the design strategy, written in a manner that teaches the theory and step by step application of the theory to solve the above-mentioned problem.
 - ✓ Add appropriate graphs and responses in your report.
 - ✓ Add proper references of the resources you used to learn and solve the problem.
 - ✓ Comment on your method of self-learning, your shortcomings and how would you improve them the next time you try to learn something yourself.

Complex Engineering Problem Attributes

WP1: Depth of knowledge WP2: Range of conflicting requirements WP3: Depth of analysis WP4: Familiarity of issues WP5: Extent of applicable codes WP6: Extent of stakeholders WP7: Interdependence	Please fill according to the WPs covered in the course CEP, example is shown here.			
	• WP1: <i>Depth of Knowledge</i> – Analysis and choice of op-amp principle for each stage requires knowledge of WK3, WK8. Design requires knowledge of WK5, WK8 • WP2: <i>Range of conflicting requirements</i> -- Requirements are conflicting. All objectives are desired, but power consumption should be minimized and DC imperfections must be dealt with. The initial amplification stage design is also critical. • WP3: <i>Depth of analysis</i> -- No unique solution. Multiple designs possible. Analysis needed to select the most appropriate one.			
	Rubrics		CLOs	Marks
	Understands objectives, requirements, and constraints	WP1, WP2	CLO 5	/ 10
	Clearly explains the controller design principles for each stage.	WP1	CLO 5	/ 10
	Accurately calculates the controller gains and associated circuit components for the design.	WP1, WP3	CLO 5	/ 10
	Develops a final design that satisfies all requirements	WP1, WP3	CLO 5	/ 10
	Design Report			/ 10
Total				/ 50

Knowledge Profiles

WK1 - Natural Sciences: A systematic theory-based understanding of natural sciences applicable to the discipline.

WK2 - Mathematics and Computing: The concept-based mathematical thinking, numerical analysis, statistics and formal aspects of computer and information science to support analysis and modelling applicable to the discipline.

WK3 - Engineering Fundamentals: A systematic, theory-based formulation of engineering fundamentals required in an engineering discipline.

WK4 - Engineering Specialization: The knowledge of Engineering specialization that provides theoretical frameworks and bodies of knowledge for the accepted practice areas that are at the forefront in a discipline.

WK5 - Engineering Design: The *Design Thinking Knowledge* that supports engineering design in a practice area of an engineering discipline.

WK6 - Engineering Practice: The Knowledge of engineering practices (technology) in different practice areas of an engineering discipline.

WK7 - Engineering in Society: A systematic, comprehension-based knowledge of the role of engineers in a society and the professional issues related to practicing engineering profession in a discipline: ethics and the professional responsibility of an engineer to public safety including the impact of an engineering activity i.e. economic, social, cultural, environmental and sustainability

WK8 - Research Literature: Engagement with selected knowledge in the research literature of the discipline.